TEAM INFORMATION

Safe Edge.

We help SDP's park teams find, fix and prove the gaps at township water edges.

People · Safety and security





Hee Zi Jie

@AlaskanTuna

Team leader · Computer science

Asia Pacific University (APU)



Yow Li Hao

@chaosiris

Computer science

University of Sheffield



Lim Yuh Kang

@kymil4

Computer science

Asia Pacific University (APU)



Liau Jin Siang

@DrxgClanPC

Law

University of Reading

1. THE PROBLEM

Storms break fences.
Nobody walks every edge.

371

child and teen drowning deaths in Malaysia, Jan 2023 to Aug 2026

Bomba, in The Star, 16 Oct 2026

8 of 21

reported township water incidents began in a drain

Our news-report table, 2023-2026; not a rate



> **“The bigger issue is … clear accountability for maintenance and public safety.”**
>
> Mr. Eric, former Head of Township, Sime Darby Property · 16 Oct 2026

2. TARGET USER

One manager.
Every edge closed.

WHO WE BUILT IT FOR

SDP's Elmina park maintenance manager

Keeps every retained water edge closed, and knows which ones are not.

- No record of where the fences are
- A weekly walk misses last night's storm
- A finished repair can leave a way round the fence end



Works with the drainage engineer, an independent reviewer and the township GM; roles to confirm with SDP

3. YOUR SOLUTION

Safe Edge: find it, fix it, prove it.



Find

A drone or the walk crew's phone spots the change. A named owner confirms it.



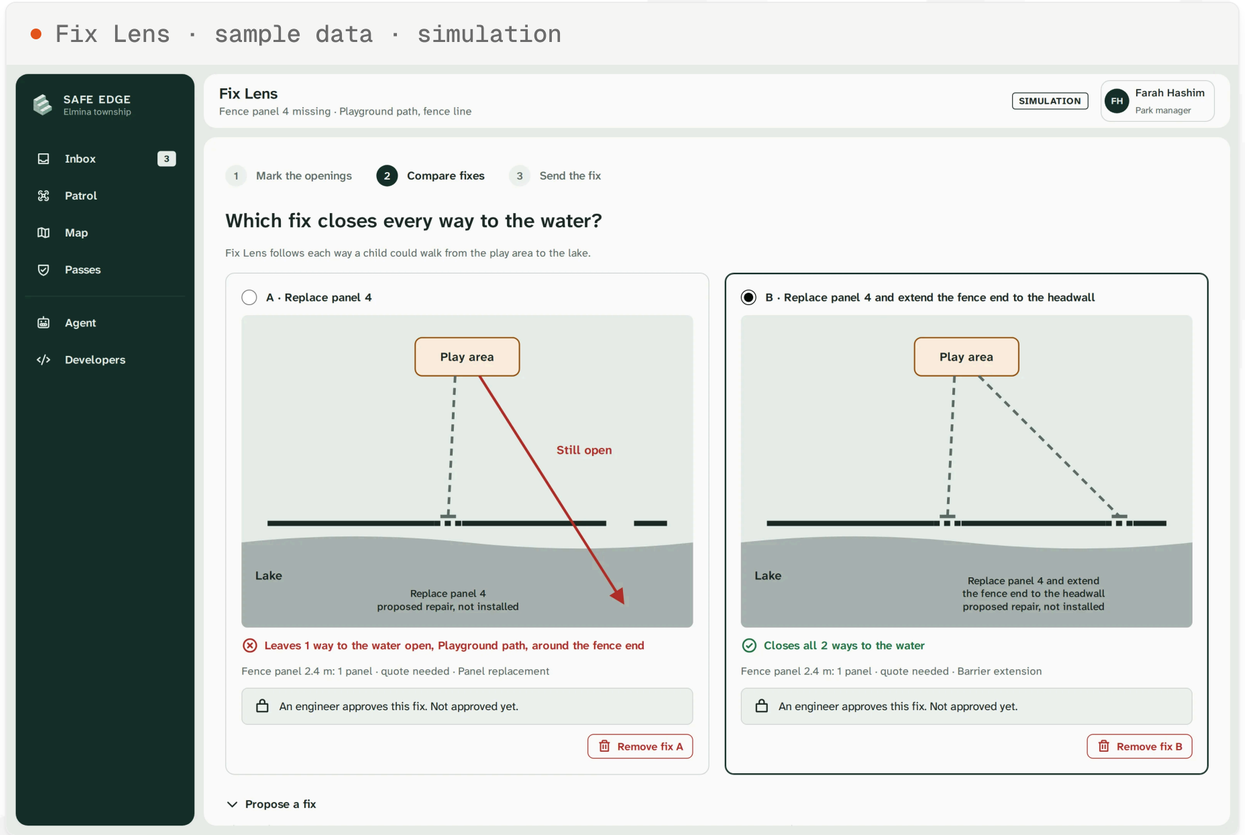
Fix

Fix Lens tests if a repair closes every way to the water, before anyone pays.



Prove

Someone who didn't install it checks on site. The edge's pass expires.



Safe Edge

4. INNOVATION

The drone is optional. The proof is the product.

Flag
AI suspends the pass

→

Named owner
A person confirms

→

Fix Lens
Every route closed?

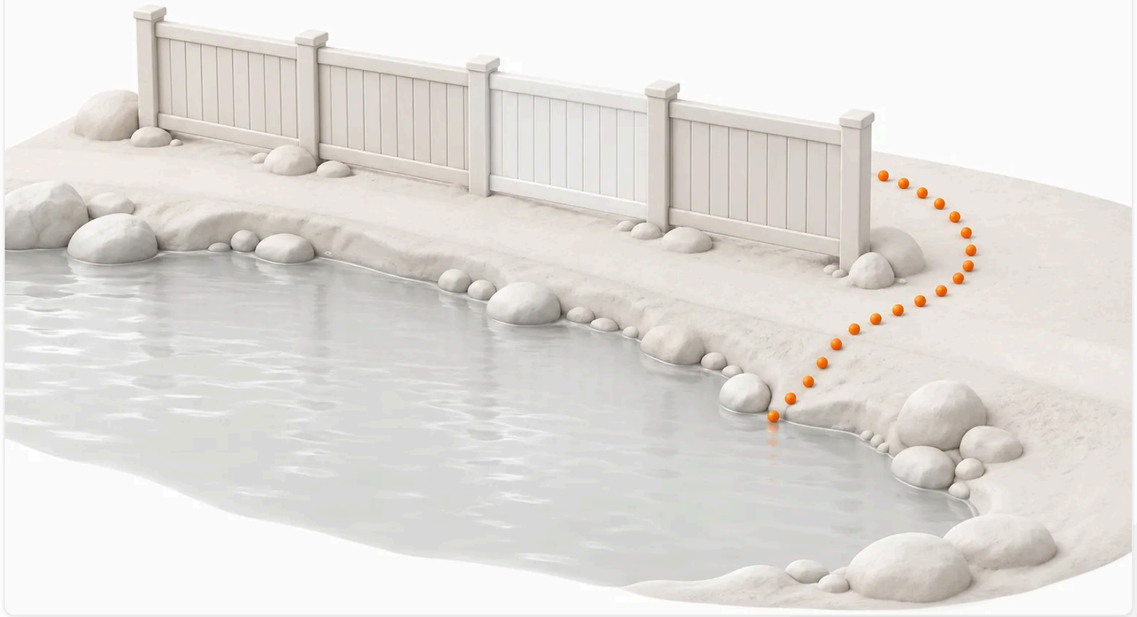
→

Independent proof
Not the installer

→

Expiry
Checked again

• A new panel, and still a way round



ALREADY EXISTS

Drone inspection and AI change detection: Singapore's PUB, DJI FlightHub 2.



OUR RULE

AI can flag and suspend. Never approve.

5. HOW IT WORKS

Six steps. Only people decide.



01

Patrol

A drone or the walk crew covers each edge

DRONE OR PHONE



02

Flag

A missing safety item suspends the pass

AI OR PATROL



03

Confirm

The manager decides in the Edge Inbox

PARK MANAGER



04

Fix

Fix Lens checks the repair closes every route

MANAGER AND ENGINEER



05

Prove

Someone who didn't install it re-tests on site

ANOTHER PERSON



06

Repeat

The pass expires; the next patrol checks again

SAFE EDGE

Unknown is never shown as safe

Safe Edge

6. IMPACT

Count the open routes. Keep them closed.

MAIN MEASURE

Day 30

Confirmed open routes to the water still closed 30 days after the repair.

Target: all of them.

Counted on site before the repair; reported with raw counts



SPEED

Days

to interim protection, and to independent acceptance

WHO GAINS

Both

Families: tested edges. **SDP:** a record for the council.

ACCESS

Open

The park stays usable. Closing it never counts as a fix.

It measures the hazard, not drownings.

SDP fit: community parks, Sustainability Report 2025, pp. 94-95

7. FEASIBILITY

It starts without a drone.

SOFTWARE

Built

Web app, API, SDK and MCP server, tested end to end

Not yet a field trial



STARTS NOW

Phone

The walk crew runs the loop with no aviation approval

Needs SDP's written access



PILOT CEILING

RM10k

for one asset; RM5,000 of it is the repair

A ceiling we set, not a quote



DRONES LATER

~2 mo

of CAAM, JUPEM, MBSA and privacy approvals, priced on top

Inside Subang's control zone; our estimate



Owner park manager · **Approves** drainage engineer · **Re-tests** independent reviewer · **Funds** township GM

Safe Edge

8. IMPLEMENTATION

One drain. Eight weeks. A gate to scale.

BEFORE WORK	**Entry gate**	Written access, a named owner and engineer, a quote, funds
WEEKS 1-2	**Baseline**	Count every route. Fix Lens against today's method
WEEKS 3-8	**Fix and prove**	Interim protection, the repair, an independent re-test
DAY 30	**Check again**	Routes still closed? Park still usable?
DECISION	**Scale gate**	Scale only if closure, access, cost and ownership hold



One retained drain, Elmina Central Park · RM10,000 ceiling · proposed; SDP has not committed

Safe Edge

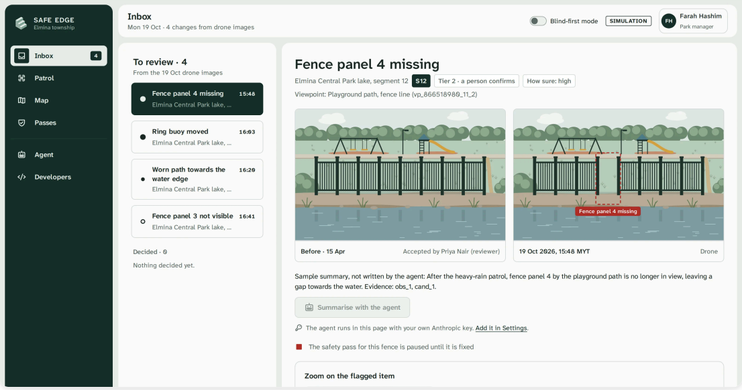
9. VISUAL CONCEPT

The working prototype, step by step.

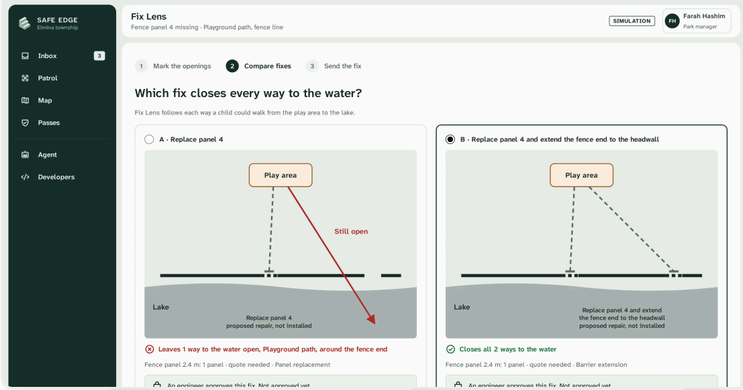
1 Find · 3D Elmina scene · simulation



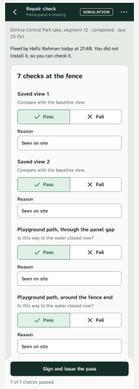
2 Confirm · Edge Inbox



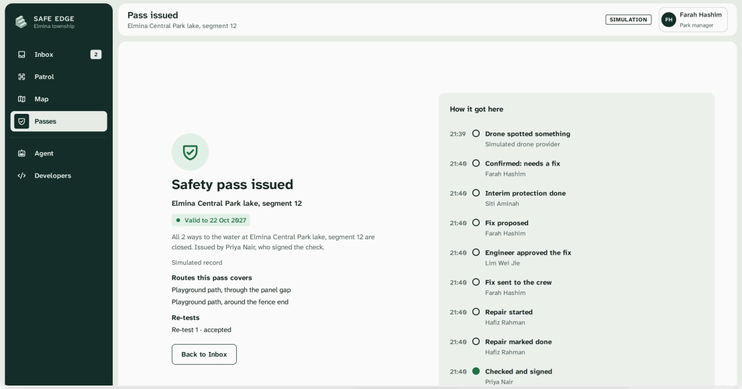
3 Fix · Fix Lens



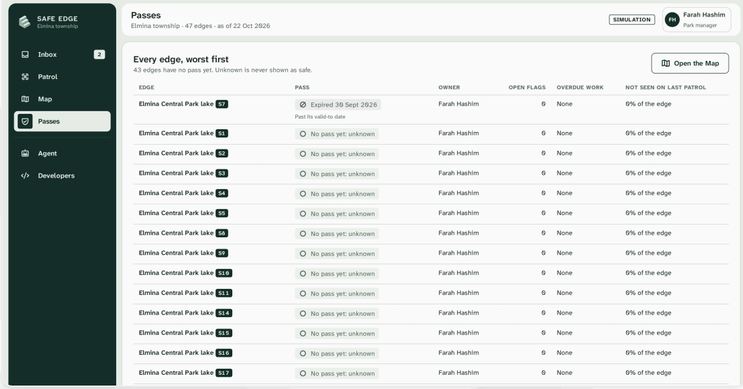
4 Prove · phone re-test



5 Pass · the record



6 Board · every edge, worst first



Live demo · safe-edge-two.vercel.app

10. WHY YOUR TEAM SHOULD WIN

We've shipped before. This one runs today.

- Three computer science students and a lawyer
- Between us: Grand Champion with Layak and with MyTakziah
- Every one of us has shipped to production, with real users
- Safe Edge runs end to end; Fix Lens catches the repair that leaves a way round

OUR ASK

An eight-week pilot on one retained drain in Elmina Central Park

A named owner · a RM10,000 ceiling

See it for yourself.

Live demo
safe-edge-two.vercel.app

Source code
github.com/TolongLabs/safe-edge

Thank you.

Safe Edge: find it, fix it, prove it.